Chunk Enrichment & Vector Databases

1. Introduction to Chunk Enrichment

Definition:
Chunk Enrichment is the systematic process of augmenting raw text chunks with additional structured and unstructured information (metadata, entities, summaries, scores, etc.) to make them more searchable, filterable, and useful for retrieval.

Why Chunk Enrichment is Critical:

- Raw chunks only contain plain text → limited semantic value during retrieval.
- Enriched chunks contain **text + intelligence** → significantly better retrieval precision and recall.
- Real-world benchmarks show **20–45% improvement** in retrieval quality when proper enrichment is applied.
- Enables advanced features like **metadata filtering**, **hybrid search**, **faceting**, and **time-based filtering**.
- Helps in compliance, security, and cost optimization.

Core Philosophy:
"Don't just store text. Store **context-aware, intelligent documents**."

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2. PII Detection & Redaction (Deep Dive)

What is PII?
Any data that can directly or indirectly identify a person.

Common PII Types (Especially Relevant in India):

- Personal: Full Name, Father's Name, Address
- Contact: Email, Phone Number, WhatsApp Number
- Government IDs: Aadhaar, PAN, Passport, Voter ID, DL
- Financial: Bank Account, UPI ID, Credit Card
- Digital: IP Address, Device ID, Geolocation

Legal & Compliance Reasons:

- **Digital Personal Data Protection (DPDP) Act 2023** – India
- GDPR (Europe), CCPA (California)
- Industry-specific: HIPAA, PCI-DSS, RBI guidelines

Detection Techniques (from basic to advanced):

Technique	Speed	Accuracy	Best Used For
Regex + Rules	Very Fast	Medium	Phone, Aadhaar, PAN, Emails
Presidio (Microsoft)	Fast	High	Production-grade redaction
spaCy + Custom	Medium	High	Domain-specific entities
LLM-based	Slow	Very High	Complex & contextual PII

Best Practice:
Redact PII **before** embedding. Store redacted version in vector DB and (optionally) original in a secure vault with access control.

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3. Named Entity Recognition (NER)

Definition:
NER is an NLP task that locates and classifies named entities in text into predefined categories.

Standard Entity Categories:

- **PERSON**, **ORG** (Organization), **GPE** (Geo-Political Entity)
- **DATE**, **TIME**, **MONEY**, **PERCENT**, **QUANTITY**
- **PRODUCT**, **EVENT**, **WORK_OF_ART**, **LAW**

Advanced Use Cases in RAG:

- Entity-based filtering (`WHERE organization = "Reliance"`)
- Building knowledge graphs (Session 4)
- Cross-document entity linking
- Domain-specific entities (Drug names in pharma, Stock symbols in finance)

Popular Tools (2025):

- Hugging Face Transformers (`dbmdz/bert-large-cased-finetuned-conll03-english`)
- spaCy (with custom models)
- LLM-powered NER (cheaper and more flexible now)

Example Output:

```
{
  "entities": [
    {"text": "Satyasai Esarapu", "label": "PERSON"},
    {"text": "Hyderabad", "label": "GPE"},
    {"text": "2025", "label": "DATE"}
  ]
}
```

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4. Key-Phrase Extraction & Chunk Summarization

Key-Phrase Extraction Methods:

- **Statistical**: TF-IDF, RAKE, YAKE
- **Embedding-based**: KeyBERT, MMR (Maximal Marginal Relevance)
- **LLM-based**: Most accurate but slower

Chunk Summarization:

- Generate 1–2 sentence summary per chunk
- Store as `summary` metadata field

Benefits:

- Improves retrieval ranking
- Useful for "summary-first" retrieval strategies
- Helps rerankers and LLMs understand chunk relevance faster

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5. Metadata Generation for Hybrid Search

What Makes Good Metadata?

- **Factual** (source, date, author)
- **Semantic** (entities, key phrases, topics)
- **Technical** (chunk_size, token_count, embedding_model)
- **Business** (department, confidence_score, access_level)

Hybrid Search Power:

- **Pre-filtering**: Apply metadata filters **before** vector search (very fast)
- **Post-filtering**: Filter after vector search
- **Combined Query Example**:
  `vector_similarity(query) AND date > "2024-01-01" AND category = "finance"`

Implementation Tip:
Store metadata as a JSON object or flattened fields in your vector DB.

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6. Vector Database Landscape (2025)

Detailed Comparison:

Vector DB	Type	Strengths	Weaknesses	Best For
Pinecone	Managed	Serverless, Excellent UX, Scale	Expensive, Vendor lock-in	Production SaaS apps
Chroma	In-process	Zero setup, Great for dev & learning	Not for production scale	Local development, Prototyping
pgvector	In-Stack	Uses existing Postgres, ACID compliant	Slower for very large scale	Teams already on PostgreSQL
Weaviate	Self-hosted/Managed	Strong schema, Hybrid search, Modules	Steeper learning curve	Complex data + Graph-like needs
Qdrant	Self-hosted	High performance, Excellent filtering	Requires DevOps	High-scale, cost-sensitive
OpenSearch	In-Stack	Full text + Vector + BM25	Heavy resource usage	Teams already using ELK stack

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7. HNSW vs IVF Indexes (Technical Deep Dive)

HNSW (Hierarchical Navigable Small World):

- Graph-based index
- Excellent recall and latency
- Memory intensive
- Default choice in most modern vector DBs

IVF (Inverted File):

- Uses clustering (k-means)
- Quantization friendly
- Better for **billions** of vectors
- Slightly lower accuracy than HNSW

Practical Recommendation:

- < 5 Million vectors → **HNSW**
- 10M+ vectors → Consider **IVF + PQ** (Product Quantization)

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8. Decision Framework – How to Choose

Decision Questions:

1. What is your current scale? (thousands vs millions of chunks)
2. Do you need serverless or full control?
3. What is your budget?
4. Are you already using any database (Postgres, Elasticsearch)?
5. Do you need advanced filtering / hybrid search?
6. Team's DevOps capability?

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9. Best Practices & Production Tips

- Redact PII as early as possible in the pipeline
- Design a consistent metadata schema from Day 1
- Start enrichment lightweight, then iterate
- Monitor storage cost vs retrieval quality
- Version your enrichment pipeline
- Use batch processing for enrichment
- Always evaluate retrieval metrics after changes

Common Pitfalls to Avoid:

- Over-enrichment (too many metadata fields)
- Storing raw sensitive data
- Ignoring index tuning (HNSW parameters)
- Using Chroma in production

Key Takeaways – Session 2

- **Chunk Enrichment** is where raw data becomes intelligent.
- PII Redaction + NER + Metadata = Foundation of production RAG.
- Metadata turns basic vector search into **Hybrid Search**.
- Choose Vector DB based on **scale, cost, control, and existing stack**.
- **Chroma** is perfect for learning. Plan for scale early.

Golden Rule:
Invest time in enrichment and storage decisions — they often yield higher returns than changing embedding models or LLMs.